MZUZU UNIVERSITY

Department of Mathematics and Statistics

Diploma in ICT

MATH 1101 - Precalculus

10th August, 2018

Time Allowed: 3 hours

INSTRUCTIONS:

- Check that the paper contains **13 questions**.
- Attempt **all** questions.
- Show all your working.

1. (a) Find the domain of $f(x) = \dfrac{\sqrt{4+x}}{1-x}$. [4]

(b) Show that $f(x) = 4x - 1$ and $g(x) = \frac{1}{4}x + \frac{1}{4}$ are inverse functions. [3]

(c) Let $f(x) = \dfrac{3x+4}{2x-5}$. Find the inverse function of $f(x)$. [4]

2. Sketch the graph of $f(x) = \dfrac{1}{x-2}$. [4]

3. Find the horizontal asymptote of $f(x) = \dfrac{3x-1}{x^2-x-6}$. [4]

4. Solve the following equation: [3]

$$\left(\frac{1}{2}\right)^{6-x} = 2.$$

5. Determine the value(s) of the constant k for which the equation has equal roots. [4]

$$x^2 + kx + 5 = 0$$

6. Find all real solutions of [8]

$$\frac{2}{\sqrt{x^2-36}}+\frac{1}{\sqrt{x+6}}-\frac{1}{\sqrt{x-6}}=0$$

7. Simplify the following expression: [6]

$$\frac{\dfrac{x}{x+2} - \dfrac{4}{x+2}}{x - 3 - \dfrac{6}{x+2}}$$

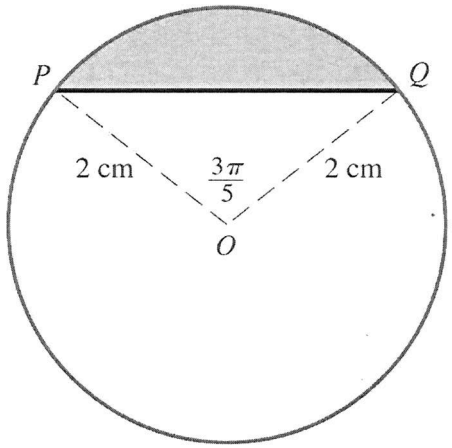
8. Solve the following inequality: [4]

$$e^{2-3x} \leq 1.$$

9. Express $\log_a \dfrac{x^3\sqrt{y}}{z^2}$ in terms of logarithms of x, y and z. [3]

10. Solve the equation $\dfrac{5^x - 5^{-x}}{2} = 3$. [8]

11. Compute the area of the segment (the shaded region) in the diagram below. Give your answer using an expression involving π. [8]



12. Prove that $\dfrac{1}{1+\sec x} + \dfrac{1}{1-\sec x} = -2\cot^2 x.$ [4]

13. Let $g(x) = -2\cos(\pi x + \frac{2\pi}{3})$.

(a) Speficy the amplitude, period and phase shift of the above function. [3]

(b) Sketch the graph of $g(x) = -2\cos(\pi x + \frac{2\pi}{3})$. [6]